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Approximate Supercell (Superlattice) Matching

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lattice; reduction; SuperCell; matching

Abstract

A solution is presented for determining the relationship between two unit cells when one is a supercell of the other.

> *Siblings are recognized at a glance, cousins on reflection, and ancestors only by argument. An acquaintance may be mistaken for any of them.*
>
> — Claude (Anthropic), 2026

1. Synopsis

A solution is proposed for the problem of best matching of a unit cell to a supercell.

2. Introduction

Santoro *et al.* (1980) enumerated transformation matrices for generating coincidence site lattices (superlattices) from a given unit cell description. They create one exact

representation of a superlattice for each choice of input and transformation. The general solution requires awareness of different choices of unit cells, alternative choices for basis vectors for actual or near degeneracies, and allowance for lattice distortion (for instance, because of replacement on one metal ion with another).

Andrews & Bernstein (2026*a*) describe a general solution for finding approximate relationships between unit cells that are similar but not necessarily sampled from the identically-same lattice. A user of that software (CmdLMP3) asked if the same could be applied to the determination of unit cells to supercells. Here we extend their method to the case of supercells.

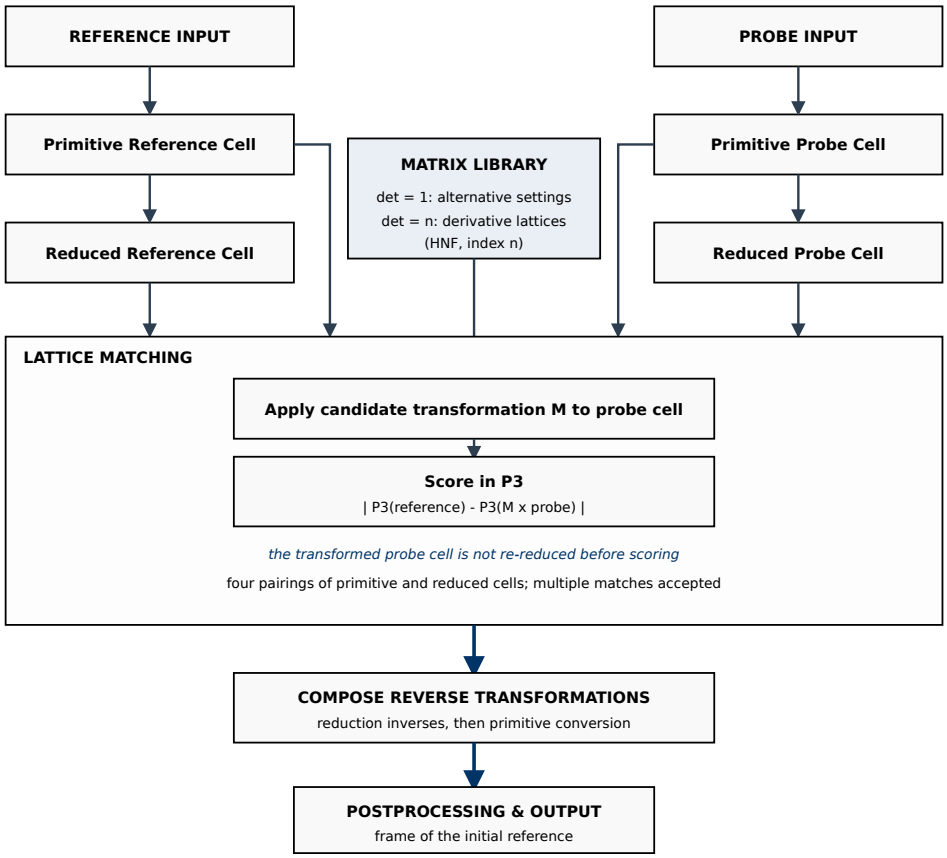


Fig. 1. The process of Andrews & Bernstein (2026) for approximate matching of lattices. Reference and probe cells are converted to primitive form and reduced. Both the primitive and the reduced cell of each are passed to the matching stage, giving four pairings. A library of matrices supplies candidate transformations of the probe: matrices of determinant one enumerate alternative settings of the probe lattice, and matrices of determinant n enumerate the derivative lattices of index n. Each candidate transformation is applied to the probe lattice and scored by the $\mathbf{P}^3$ distance to the reference. The transformed cell is not reduced again before scoring, so the score depends on the setting as well as on the lattice. Accepted matches are carried back through the reduction and primitive-conversion inverses to the frame of the initial reference.

3. Algorithm

The above algorithm suffices for matching lattices that are somewhat similar to each other, but the question of matching a cell to a supercell can be tackled with a modest modification. Where matching above is done using unimodular matrices, by replacing the library of transformation matrices with a library of matrices that have larger determinants, approximate supercell relationships can be determined.

For instance, for a given unit cell, to find supercells that have a base vector tripled a library of transformations with determinant=3.0 would be used. Other than that, the algorithm would only need output reporting changed.

4. Notation

$\mathbf{P}^3$: Polar coordinate space is a compact and smooth alternative to lengths and angles. $\mathbf{P}^3$ is useful for comparing unit cells. (Andrews & Bernstein, 2026*b*) $\mathbf{P}^3$:

$(|\vec{a}|, \alpha),\ (|\vec{b}|, \beta),\ (|\vec{c}|, \gamma)$

$\Rightarrow$

$(|\vec{a}| \cos\alpha,\ |\vec{a}| \sin\alpha)$

$(|\vec{b}| \cos\beta,\ |\vec{b}| \sin\beta)$

$(|\vec{c}| \cos\gamma,\ |\vec{c}| \sin\gamma)$

also known as (p_1, p_2, p_3)

5. Implementation details

5.1. Matrix generator

Appendix A is a C^{++} function for generating libraries of matrices with specified integer valued determinants. In program CmdLMP3, this function is used to create the library of matrices that are used for approximate lattice matching.

5.2. Determinants

For each value of determinants (n), the number of possible matrices with maximum elements (-n,n), the number of possible matrices grows rapidly. See Table 1.

Table 1. *Transformation matrix counts by determinant class. For* $n = 1$ *and* $n = 2$ *the complete set is used in CmdLMP3. For* $n = 3$ *and* $n = 4$ *the complete set is used with the 13 and 35 Hermite normal forms appended, respectively; those matrices are already present in the complete set, so the appended copies are redundant and are retained only as a guard. For index* $n = 5$ *and index* $n = 6$ *the exhaustive search is not attempted, as it would screen* 2.4×10^9 *and* 1.1×10^{10} *candidates to yield sets of* 1.4×10^7 *and* 7.1×10^7 *matrices. In their place the 3480 matrices of determinant one are composed with the Hermite normal forms of index 5 and 6 in both right and left multiplication orders. Left multiplication preserves the lattice and supplies alternative settings of it; right multiplication changes the lattice. The resulting sets span all 31 and all 91 derivative lattices, but sample the settings of each lattice less densely than the exhaustive sets at lower index.*

n	Derivative lattices†	Candidates screened‡	Complete set§	Used in CmdLMP3¶
1	1	19 683	3 480	3 480
2	7	1 953 125	147 696	147 696
3	13	40 353 607	1 060 920	1 060 933
4	35	387 420 489	6 721 896	6 721 931
5	31	2 357 947 691	13 942 248	215 760
6	91	10 604 499 373	70 632 144	633 360
Total	177		92 508 384	8 783 160

† Number of distinct sublattices of index n in a three-dimensional lattice. Each has exactly one Hermite normal form. The total excludes the trivial lattice at $n = 1$.

‡ Size of the exhaustive search over integer matrices with coefficients in $[-n, n]$, the smallest symmetric box that contains every Hermite normal form of index n. Screening costs are not additive across independent searches, so no total is given.

§ Number of those candidates having determinant $+n$. Matrices of negative determinant are excluded because they invert chirality.

¶ Number of matrices actually retained by CmdLMP3.

5.3. Reference vs. probe decision

For a given input pair of unit cells, the decision to search for supercells of the reference cell requires that the reference have the smaller primitive unit volume of the pair. The output of the results of matching shows the transformation from the input probe cell to the frame of the input reference cells. That might be the reverse of the

choice that the user desires, so the inverse 3x3 matrix is also output.

6. Examples

6.1. Example 1

```
=== TEST: Hawaiian ===
Description: Nearly identical cells with known transformation
Input cells:
Cell 1 (P):    10.250    10.740    21.080    87.720    75.970    61.530
Cell 2 (P):    10.250    10.740    21.080    78.960    75.970    61.490
; primitive cell volume ratio, mobile/reference
;  ratio = 1.000
; Found 403 results

; === LATTICE MATCHING RESULTS ===

; --- Match 1 ---
; EQUIVALENT  P3 distance: 0.010 (0.0% of P3 norm)  S6 angle: 0.01 deg
; Matrix:  [-1 0 0   -1 1 0   0 0 -1] (Determinant: 1.000)
; Inverse: [-1 0 0   -1 1 0   0 0 -1]
; P    10.250    10.740    21.080    87.720    75.970    61.530  [REFERENCE]
; P    10.250    10.739    21.080    87.714    75.970    61.502  [TRANSFORMED MOBILE (CONVENTIONAL)]

; --- Match 2 ---
; EQUIVALENT  P3 distance: 0.023 (0.1% of P3 norm)  S6 angle: 0.03 deg
; Matrix:  [-1 0 0   0 -1 0   -1 0 1] (Determinant: 1.000)
; Inverse: [-1 0 0   0 -1 0   -1 0 1]
; P    10.250    10.740    21.080    87.720    75.970    61.530  [REFERENCE]
; P    10.250    10.740    21.087    87.674    75.893    61.490  [TRANSFORMED MOBILE (CONVENTIONAL)]
```

6.2. Example 2

```
 === TEST: FaceDiagonalOrder2 ===
 Description: [SUPERCELL] Cubic doubled along face diagonal -- order-2, dist=0
 Input cells:
 Cell 1 (P):    10.000    10.000    10.000    90.000    90.000    90.000
 Cell 2 (P):    14.142    14.142    10.000    90.000    90.000    90.000
 ; primitive cell volume ratio, mobile/reference
 ;  ratio = 2.000   swapped reference and probe cells
 ; Found 324 results

 ; === LATTICE MATCHING RESULTS ===
 ; === SUPERCELL ===  prim. cell volume ratio = 2.000

 ; --- Match 1 ---
 ; order-2 supercell  EXCELLENT  P3 distance: 0.000 (0.0% of P3 norm)  S6 angle: 0.00 deg
 ; Matrix:  [1 -1 0   1 1 0   0 0 1] (Determinant: 2.000  [SUPERCELL])
 ; Inverse: [1/2 1/2 0   -1/2 1/2 0   0 0 1]
 ; P    14.142    14.142    10.000    90.000    90.000    90.000  [REFERENCE]
 ; P    14.142    14.142    10.000    90.000    90.000    90.000  [TRANSFORMED MOBILE (CONVENTIONAL)]

 ; --- Match 2 ---
```

```
; order-2 supercell  EXCELLENT  P3 distance: 0.000 (0.0% of P3 norm)  S6 angle: 0.00 deg
; Matrix:  [1 1 0   -1 1 0   0 0 1] (Determinant: 2.000  [SUPERCELL])
; Inverse: [1/2 -1/2 0   1/2 1/2 0   0 0 1]
; P    14.142    14.142    10.000    90.000    90.000    90.000  [REFERENCE]
; P    14.142    14.142    10.000    90.000    90.000    90.000  [TRANSFORMED MOBILE (CONVENTIONAL)]
```

6.3. Example 3

```
=== TEST: Simon Parsons ===
Description: [SUPERCELL] R primitive vs P triclinic -- order-4, dist=0.820
Input cells:
Cell 1 (P):     9.477     9.477     9.477    64.240    64.240    64.240
Cell 2 (P):     9.482    13.705    20.048    89.510    88.720    86.960
; primitive cell volume ratio, mobile/reference
;  ratio = 3.953  swapped reference and probe cells
; Found 300 results

; === LATTICE MATCHING RESULTS ===
; === SUPERCELL ===  prim. cell volume ratio = 4.000

; --- Match 1 ---
; order-4 supercell  POOR  P3 distance: 0.820 (3.1% of P3 norm)  S6 angle: 1.16 deg
; Matrix:  [0 1 0   -1 1 -1   -2 0 2] (Determinant: 4.000  [SUPERCELL])
; Inverse: [1/2 -1/2 -1/4   1 0 0   1/2 -1/2 1/4]
; NOTE: P3 distance (3.146% of P3 norm) exceeds threshold 2.000% -- match is distorted
; P     9.482    13.705    20.048    89.510    88.720    86.960  [REFERENCE]
; P     9.477    13.834    20.155    90.000    90.000    84.859  [TRANSFORMED MOBILE]

Validating using quality thresholds:
Checking result 1 ([SUPERCELL] order-4 det~4 dist~1.9):
```

7. Summary

A method for determining whether a reference cell and a probe cell have a subcell/supercell relationship and for determining the three-dimensional transformation matrices between them is described.

The described algorithm (CmdLMP3) on a modest laptop computer processes det=1 cases in about one second. Det=4 cases take the order of one minute.

8. Availability of code

The code for Lattice Matching (CmdLMP3) and related software tools is available in github.com, in `https://github.com/duck10/LatticeRepLib.git`. The program

CmdLMP3 uses the required files.

9. Supporting information

CmdLP3 has two functions in addition to the processing of users' inputs. Their results are included as supporting information.

Two text files are deposited. `CmdLMP_HELP_2026_07_18.txt` contains descriptions of possible input parameters that can be used in addition to experimental data input. For instance, "DETAILS" will trigger output of information about intermediate calculations. `CmdLMP3_TEST_2026_07_18.txt` contains about 50 example cases that the command "TEST" causes to be executed.

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Appendix A
Supercell generation code

This is a C++ function to generate all 3x3 matrices with a specified determinant value and the range of coefficients from -maxCoeff to maxCoeff.

```
inline std::vector<Matrix_3x3> generateSupercellMatrices(
         const std::vector<int>& determinants,
         const int maxCoeff = 1) {

 std::vector<Matrix_3x3> matrices;
 matrices.reserve(8000);

 std::set<int> detSet(determinants.begin(), determinants.end());

 const int lo = -maxCoeff;
 const int hi = maxCoeff;

 for (int a00 = lo; a00 <= hi; ++a00)
 for (int a01 = lo; a01 <= hi; ++a01)
 for (int a02 = lo; a02 <= hi; ++a02)
 for (int a10 = lo; a10 <= hi; ++a10)
 for (int a11 = lo; a11 <= hi; ++a11)
 for (int a12 = lo; a12 <= hi; ++a12)
 for (int a20 = lo; a20 <= hi; ++a20)
 for (int a21 = lo; a21 <= hi; ++a21)
 for (int a22 = lo; a22 <= hi; ++a22) {
  Matrix_3x3 M(
  a00, a01, a02,
  a10, a11, a12,
  a20, a21, a22
 );
 const int det =
 static_cast<int>(std::round(M.Det()));
 if (detSet.count(det))
    matrices.emplace_back(M);
 }

 matrices.shrink_to_fit();
 return matrices;
}
```

References

Andrews, L. C. & Bernstein, H. J. (2026*a*). *Foundations of Crystallography*, **82**(2).
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Santoro, A., Mighell, A. & Rodgers, J. (1980). *Foundations of Crystallography*, **36**(5), 796–800.